

PDA College Of Engineering
Department Of Computer Science and Engineering

SYNOPSIS OF THE PROPOSED MAJORPROJECT (22CS65)

ON

AirSketch: Gesture Based Virtual Drawing Platform

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Department Vision Mission Statement

Vision

To become a premier department in Computer education, research and to prepare highly competent IT professionals to serve industry and society at local and global levels.

Mission

- To impart high quality professional education to become a leader in Computer Science and Engineering.
- To achieve excellence in research for contributing to the development of the society.
- To inculcate professional and ethical behaviour to serve the industry.

ABSTRACT

The rapid advancement of computer vision and artificial intelligence has enabled new ways for humans to interact with computers beyond traditional input devices such as keyboards, mice, and touch screens. Gesture-based interaction has emerged as an intuitive and natural method of communication between humans and machines. By recognizing hand movements and gestures through cameras, computer systems can interpret user actions and translate them into digital commands. This technology has gained significant attention in fields such as virtual reality, smart interfaces, and interactive applications where touchless interaction improves usability and accessibility.

This project proposes the development of a “**AirSketch: Gesture Based Virtual Drawing Platform**”, an interactive web application that allows users to draw on a digital canvas using hand gestures detected through a webcam. The system uses modern computer vision technologies to track hand landmarks and interpret finger movements in real time. Frameworks such as MediaPipe and TensorFlow.js enable accurate detection of hand gestures by analyzing multiple key points of the hand and identifying patterns that correspond to specific commands. By converting these gestures into drawing actions, the system eliminates the need for physical input devices while providing a smooth and responsive drawing experience.

The proposed system will support various gesture-based controls to perform different functions on the canvas. Users will be able to draw by extending a single finger, change colors by performing specific gestures, switch to eraser mode, and clear the canvas using predefined hand movements. The application will include a user-friendly interface with options for selecting colors and adjusting brush size, making the drawing experience more flexible and interactive. Real-time hand tracking will ensure that user gestures are accurately interpreted and immediately reflected on the canvas.

The expected outcome of this project is a functional gesture-controlled drawing platform that demonstrates how computer vision and artificial intelligence can be used to create intuitive human–computer interaction systems. By integrating real-time hand tracking with a responsive web interface, the AirSketch aims to showcase the potential of touchless input technologies in modern digital applications and provide a foundation for future developments in gesture-driven interfaces.

INTRODUCTION

Human-computer interaction has traditionally relied on physical input devices such as keyboards, mice, styluses, and touch screens. While these devices are effective for most computing tasks, they require direct physical contact and sometimes limit the natural way humans interact with digital systems. With the rapid development of computer vision and artificial intelligence technologies, researchers and developers are exploring more intuitive interaction methods that allow users to control systems through natural movements and gestures. Gesture recognition has therefore become an important area of research in modern computing, enabling machines to understand human actions through visual input captured by cameras.

Hand gesture recognition is one of the most widely studied approaches in this field. By detecting and analyzing the movement and position of fingers and hands, computer systems can interpret specific gestures as commands. This technology has been used in various applications such as virtual reality environments, interactive gaming systems, smart classrooms, sign language recognition, and touchless control interfaces. Gesture-based systems offer several advantages, including improved accessibility, reduced dependence on physical devices, and a more engaging user experience.

One practical application of gesture recognition technology is the development of virtual drawing systems. Digital drawing platforms are commonly used in education, design, and creative fields, but they usually depend on tools such as a mouse, graphics tablet, or touchscreen stylus. These tools may not always be available or convenient for all users. A gesture-based drawing system allows users to create digital artwork simply by moving their hands in front of a camera, making the process more natural and interactive. Such systems can also be useful in situations where touchless interaction is preferred, such as public displays, collaborative environments, or immersive digital experiences.

The project “**AirSketch: Gesture Based Virtual Drawing**” aims to develop an interactive web application that enables users to draw on a digital canvas using hand gestures detected through a webcam. The system will track hand movements, identify specific finger configurations, and translate these gestures into drawing actions such as creating lines, selecting colors, erasing strokes, and clearing the canvas. By combining real-time hand tracking with a responsive user interface, the platform will provide a smooth and intuitive drawing experience without relying on traditional input devices.

Overall, **AirSketch** explores how computer vision and gesture recognition technologies can enhance human-computer interaction. By enabling users to control a digital canvas through natural hand movements, the system aims to create a more intuitive and engaging interface while demonstrating the practical applications of AI-driven gesture recognition in interactive software systems.

EXISTING AND PROPOSED SOLUTIONS

Existing Solutions:

• AirCanvas — Gesture-Based Drawing System:

AirCanvas is a computer vision–based drawing application that allows users to draw in the air using hand gestures captured through a webcam. The system detects the position of the user’s hand and tracks finger movement to create drawings on a digital canvas. Most implementations use Python, OpenCV, and MediaPipe to detect hand landmarks and convert them into drawing coordinates.

• Gesture Controlled Virtual Mouse:

Gesture Controlled Virtual Mouse systems allow users to control the mouse pointer using hand gestures detected through a camera. These systems interpret different finger positions as commands such as clicking, dragging, or scrolling. Technologies such as computer vision, OpenCV, and machine learning are commonly used to track hand landmarks and recognize gestures.

Proposed Solution:

AirSketch aims to develop an interactive web-based platform that allows users to draw on a digital canvas using hand gestures detected through a webcam. The system utilizes modern computer vision frameworks such as MediaPipe and TensorFlow.js to detect hand landmarks and recognize finger gestures in real time directly within a web browser.

By tracking finger movements, the system converts gestures into drawing commands such as creating lines, selecting colors, switching to eraser mode, and clearing the canvas. Unlike many existing systems that are limited to desktop environments or simple gesture detection, the proposed platform provides a responsive web interface with multiple gesture controls and real-time visual feedback which enhances user interaction and accessibility.

PROBLEM STATEMENT

Traditional digital drawing systems rely on physical input devices such as a mouse, stylus, or touchscreen, which may not always provide the most natural or intuitive way for users to interact with a computer. These devices require direct physical contact and can limit flexibility in situations where touchless interaction is preferred.

Additionally, many existing gesture-based systems focus mainly on basic cursor control or simple demonstrations of hand tracking rather than providing a complete and interactive application for practical use. As a result, there is a need for a system that can effectively utilize computer vision and hand gesture recognition to create an intuitive and interactive digital drawing environment.

METHODOLOGY

1. Video Input and Data Capture:

The first stage of the system involves capturing real-time video input using the user's webcam. The camera continuously records frames and sends them to the application for processing. These video frames act as the primary data source for detecting hand movements and gestures that will later be translated into drawing commands on the digital canvas.

2. Hand Detection and Landmark Identification:

In this stage, computer vision frameworks such as MediaPipe are used to detect the user's hand within each video frame. The system identifies multiple key landmark points on the hand, including finger joints and fingertips. These landmarks help the system understand the position, orientation, and movement of the hand in real time.

3. Gesture Recognition:

Once the hand landmarks are detected, the system analyzes the position and configuration of the fingers to recognize specific gestures. This stage interprets the user's hand posture and converts it into meaningful control signals.

4. Coordinate Mapping and Drawing Logic:

After recognizing the gesture, the system maps the fingertip position detected in the camera frame to the coordinates of the digital canvas. When the system detects a drawing gesture, the movement of the fingertip is translated into continuous lines on the canvas.

5. Canvas Interaction and Tool Selection:

The application includes a user interface that allows interaction with different drawing tools such as color selection, brush size adjustment, and eraser mode. Specific gestures trigger these functions, enabling users to control the canvas without using traditional input devices.

6. Web Application Development:

The entire system is implemented as a web-based application using modern frontend technologies such as React, TypeScript, and Tailwind CSS. AI libraries such as TensorFlow.js are integrated into the browser environment to process gesture detection in real time.

7. System Testing and Performance Evaluation:

The final stage involves testing the system to ensure accurate gesture detection, smooth drawing performance, and reliable user interaction. The application is evaluated under different lighting conditions and user movements to verify that the gesture recognition system performs consistently and provides a responsive drawing experience.

SYSTEM REQUIREMENTS

1. Hardware Requirements

- **Processor:** Intel i3 or higher
- **RAM:** Minimum 4 GB (8 GB recommended)
- **Storage:** At least 10 GB free disk space
- **Camera:** Webcam for capturing hand gestures
- **Internet Connection:** Required for installing dependencies and running web frameworks
- **Input Devices:** Keyboard and mouse for system control and development
- **Output Devices:** Monitor or display screen

2. Software Requirements

- **Operating System:** Windows / macOS / Linux
- **Frontend Technologies:** HTML, CSS, JavaScript
- **Frameworks/Libraries:** React, Tailwind CSS, Vite
- **Programming Language:** TypeScript / JavaScript
- **AI and Computer Vision Tools:** MediaPipe, TensorFlow.js
- **Development Tools:** Node.js, npm or yarn package manager
- **Web Browser:** Google Chrome / Microsoft Edge / Firefox (with webcam access)

SYSTEM SPECIFICATIONS

Reliability:

The system should accurately detect hand gestures and translate them into drawing commands with minimal errors.

Performance:

Gesture recognition and canvas updates should occur in real time to ensure a smooth and responsive drawing experience.

Usability:

The application interface should be simple and intuitive so that users can easily perform drawing operations using hand gestures.

Scalability:

The system should allow the addition of new gestures, drawing tools, and features without affecting the existing functionality.

Security:

The application should ensure safe access to the webcam and protect user privacy by processing gesture data locally within the browser.

EXPECTED OUTCOME OF THE PROJECT

The expected outcome of the proposed project is the development of a functional and interactive web application that allows users to draw on a digital canvas using hand gestures captured through a webcam. The system will demonstrate how computer vision and artificial intelligence technologies can be used to create intuitive human–computer interaction methods that do not rely on traditional input devices such as a mouse or stylus. By using real-time hand tracking, the platform will allow users to perform drawing actions simply by moving their fingers in front of the camera.

One of the key outcomes of the system will be the successful implementation of gesture recognition for controlling different canvas operations. The application will detect specific finger configurations and interpret them as commands such as drawing lines, selecting colors, switching to eraser mode, and clearing the canvas. This gesture-based control mechanism will provide a smooth and natural drawing experience where the movement of the user’s finger directly corresponds to the strokes drawn on the screen.

Another important outcome of the project will be the integration of computer vision frameworks and web technologies into a single interactive platform. By using tools such as MediaPipe and TensorFlow.js, the system will be capable of detecting hand landmarks and tracking finger movements in real time within a web browser. The web application interface will provide users with essential drawing tools, color options, and visual feedback to improve usability and interaction.

The system will also demonstrate the potential of touchless interfaces in modern computing environments. Gesture-controlled systems can be useful in various fields such as education, digital art, virtual collaboration, and interactive displays. The project will highlight how natural hand movements can be used as an alternative input method, making digital interaction more engaging and accessible.

Overall, the **“AirSketch: Gesture Based Virtual Drawing”** project aims to showcase how artificial intelligence and computer vision can enhance user interaction with digital systems. The completed platform will serve as a working prototype that illustrates the practical applications of gesture recognition in web-based software. In addition to providing an innovative drawing experience, the project will act as a foundation for future developments in gesture-controlled applications and AI-powered interactive systems.